Top Interview Master Questions & Learning Roadmap

Arrays & Hashing

Master Questions:

- Two Sum
- Longest Consecutive Sequence
- Product of Array Except Self
- Top K Frequent Elements
- Subarray Sum Equals K

Core Concepts:

- HashMap
- Prefix Sum
- Frequency Counting

Sliding Window

Master Questions:

- Longest Substring Without Repeating Characters
- Minimum Window Substring
- Permutation in String
- Fruits Into Baskets

Core Concepts:

- Fixed Window
- Dynamic Window
- Frequency Map

Two Pointer

Master Questions:

- Container With Most Water
- 3Sum
- Trapping Rain Water
- Move Zeroes

Core Concepts:

- Opposite Direction Pointers

- Fast-Slow Pointer

Binary Search

Master Questions:

- Search in Rotated Sorted Array
- Median of Two Sorted Arrays
- Koko Eating Bananas
- Find Peak Element

Core Concepts:

- Binary Search on Answer
- Monotonic Predicate

Stack

Master Questions:

- Valid Parentheses
- Daily Temperatures
- Largest Rectangle in Histogram
- Next Greater Element

Core Concepts:

- Monotonic Stack
- Expression Evaluation

Linked List

Master Questions:

- Reverse Linked List
- Merge Two Sorted Lists
- LRU Cache
- Detect Cycle

Core Concepts:

- Fast/Slow Pointer
- Dummy Node

Trees & BST

Master Questions:

- LCA in Binary Tree
- Diameter of Binary Tree
- Validate BST
- Serialize Deserialize Tree

Core Concepts:

- DFS
- BFS
- Tree DP

Graphs

Master Questions:

- Number of Islands
- Course Schedule
- Word Ladder
- Alien Dictionary

Core Concepts:

- BFS
- DFS
- Topological Sort
- Union Find

Dynamic Programming

Master Questions:

- Climbing Stairs
- House Robber
- Coin Change
- Edit Distance

Core Concepts:

- Memoization
- Tabulation
- State Transition

Greedy

Master Questions:

- Jump Game
- Gas Station
- Merge Intervals

Core Concepts:

- Sorting Based Greedy
- Local Optimum

Backtracking

Master Questions:

- N Queens
- Sudoku Solver
- Combination Sum

Core Concepts:

- Choose → Explore → Unchoose

Advanced Topics

Master Questions:

- Segment Tree
- DP on Trees
- Trie + DFS
- Concurrent LRU Cache

Core Concepts:

- Range Query
- Lazy Propagation
- Concurrency

Best Learning Resources

Platform	Purpose
LeetCode	Practice interview problems
NeetCode	Structured interview roadmap
TakeUForward	DSA sheets and explanations
William Fiset	Advanced graph and algorithm concepts
Abdul Bari	Core algorithm fundamentals
Codeforces EDU	Advanced competitive programming concepts

Recommended Study Order

1. Phase 1: Arrays → Hashing → Sliding Window → Two Pointer
2. Phase 2: Stack → Linked List → Binary Search → Heap
3. Phase 3: Trees → BST → Graphs
4. Phase 4: Dynamic Programming → Trie → Backtracking
5. Phase 5: Advanced DP → Segment Tree → Union Find